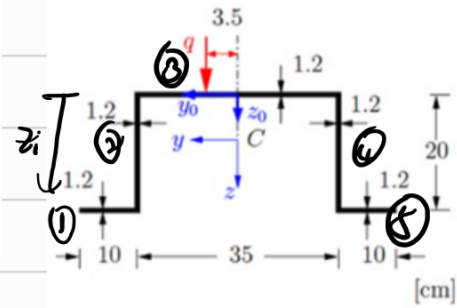


c) the distribution of the shear stresses due to shear force and torsion across the profile. Determine position and value of the largest shear stress..



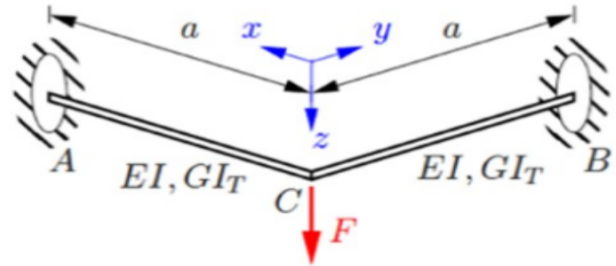
$$I_1 = \frac{M_1}{W_1} = \frac{4.2 \times 10^2}{48.6} = 9.21 \text{ kN/cm}^2$$

$$c) \tau_{\text{inside}} = 27.6 - 9.21 = -6.45 \text{ kN/cm}^2$$

$$\tau_{\text{outside}} = 27.6 + 9.21 = 11.97 \text{ kN/cm}^2$$

$$\tau_{\text{max}} = 11.97 \text{ kN/cm}^2$$

Homework 5.6 The depicted cantilever is fixed at both ends and bent by 90° . The cantilever is loaded at point C by the force F. Compute the deflection at the point C.
Hint: The rotation at C must be compatible for both cantilevers.



$$M_y = -\frac{F}{2}(a-l) + M$$

$$EI w'' = -M_y = \frac{F}{2}(a-l) - M$$

$$EI w' = \frac{F}{2}\left(al - \frac{l^2}{2}\right) - Ml + C_1$$

$$EI w = \frac{F}{2}\left(al^2 - \frac{l^3}{6}\right) - \frac{Ml^2}{2} + C_1 l + C_2$$

$$w_C(a) = \frac{1}{EI} \left[\frac{F}{2} \left(\frac{a^3}{2} - \frac{a^3}{6} \right) - \frac{Ma^2}{2} \right] = \frac{1}{EI} \left[\frac{Fa^3}{6} - \frac{Ma^2}{2} \right]$$

$$w'_{A,C} = w'(a) = \frac{1}{EI} \left[\frac{F}{2} \left(a^2 - \frac{a^2}{2} \right) - Ma \right]$$

$$= \frac{1}{EI} \left[\frac{Fa^2}{4} - Ma \right]$$

$$w'_{A,C} = \theta_{B,C}$$

$$\theta_C = \frac{Ma}{GI_T} = \frac{1}{EI} \left[\frac{Fa^2}{4} - Ma \right]$$

$$Ma \left[\frac{1}{EI} + \frac{1}{GI_T} \right] = \frac{Fa^2}{4EI}$$

$$M = \frac{Fa}{4EI} \times \frac{EI \cdot GI_T}{GI_T + EI}$$

$$= \frac{Fa}{4} \times \frac{GI_T}{EI + GI_T}$$

$$w'(0) = 0 \Rightarrow C_1 = 0$$

$$w(0) = 0 \Rightarrow C_2 = 0$$

$$\begin{aligned}
 \omega_c &= \frac{1}{EI} \left[\frac{Fa^3}{6} - \frac{Ma^2}{6} \right] = \frac{1}{EI} \left[\frac{Fa^3}{6} - \frac{Fa^3}{4} \left(\frac{GI_T}{EI + GI_T} \right) \right] \\
 &= \frac{Fa^3}{EI} \left[\frac{1}{6} - \frac{GI_T}{4(EI + GI_T)} \right] \\
 &= \frac{Fa^3}{24EI} \left[\frac{4EI + GI_T}{EI + GI_T} \right]
 \end{aligned}$$